MARANDA 2025 KCSE PREDICTION

EXPECTED EXAM 1

CHEMISTRY

233/3

PAPER 3 (PRACTICAL)

TIME: 2¼ HOURS

NAME.....

SCHOOL..... **SIGN**.....

INDEX NO..... **ADM NO**.....

Kenya Certificate of Secondary Education.

INSTRUCTIONS TO CANDIDATES

- *Write your name in the spaces provided above.*
- *Answer all the questions in the spaces provided*
- *All workings must be clearly shown where necessary*

FOR EXAMINERS USE ONLY

QUESTION	MAXIMUM SCORE	CANDIDATE'S SCORES
1	20	
2	11	
3	09	
TOTAL SCORE	**40**	

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QUESTIONS

1. You are provided with:

- 0.30g solid A, magnesium metal
- Hydrochloric acid, solution B
- 0.15M sodium carbonate, solution C
- Methyl orange indicator

You are required to determine:

- The enthalpy change, ΔH per mole for the reaction between magnesium and excess hydrochloric acid.
- The concentration in moles per litre of hydrochloric acid, solution B.

PROCEDURE I

i. Using a burette, place 50.0cm³ of Solution B in a 100 ml plastic beaker.

ii. Measure the temperature of solution B after every 30 seconds. Record your readings in Table 1.

iii. At the 90th second, add **ALL** of solid A provided to the beaker, stir gently with the thermometer and continue measuring and recording the temperature after every 30 seconds. Complete Table

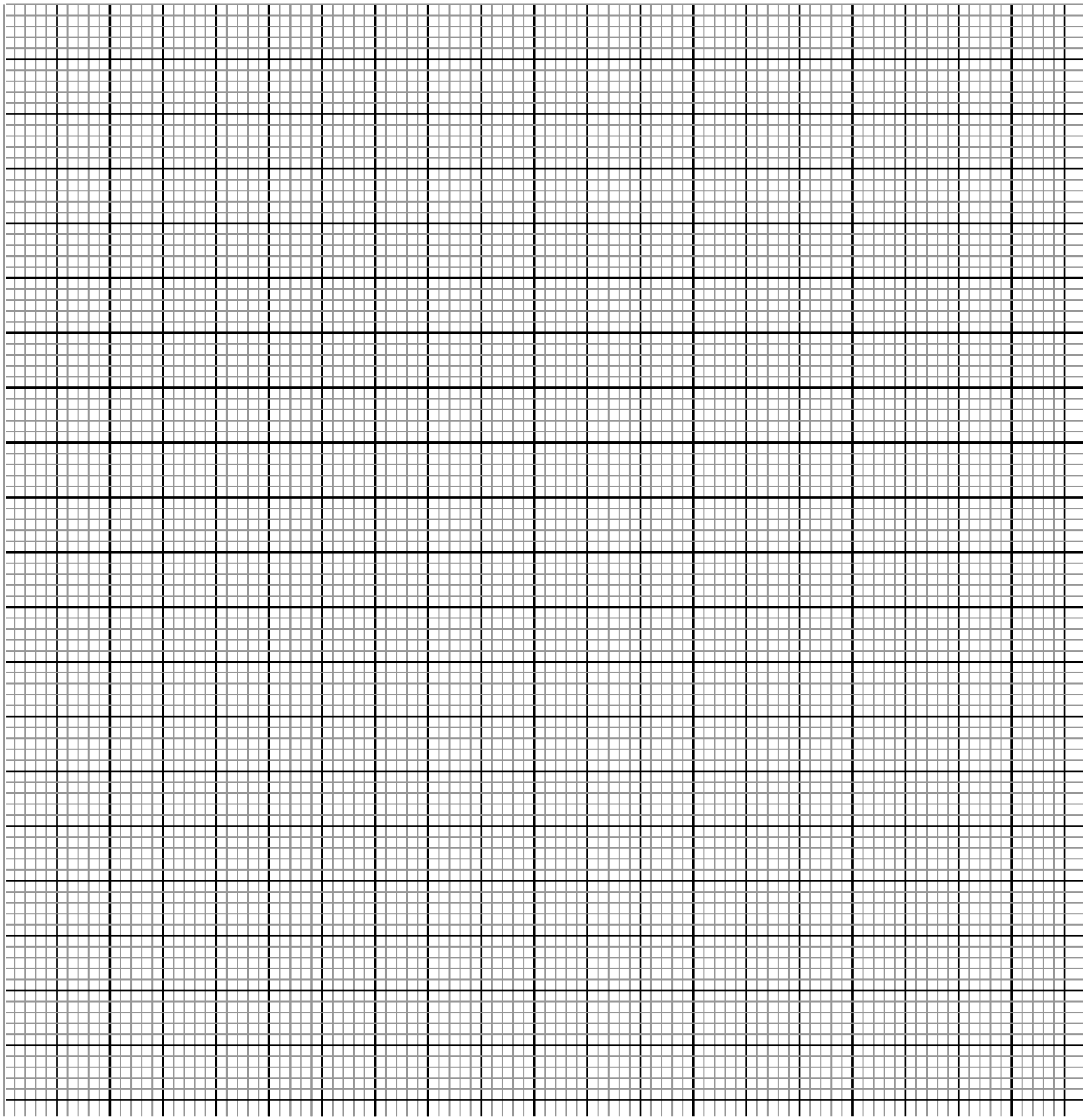
1. **Retain the content in the beaker for use in procedure II**.

Table 1

Time (S)	0	30	60	90	120	150	180	210	240	270
emperature (°C)				X						

(3 marks)

a) On the grid provided, plot a graph of temperature (vertical axis) against time. **(3 marks)**



b) Determine the change in temperature, ΔT, for the complete reaction. Show the working on the graph.

ΔT=…… **(1mark)**

c) Calculate the heat change for the reaction. *(Assume density of solution =1.0g/cm ^{3}, specific capacity of the solution = 4.2 kJ g^{-1} K^{-1}).* **(1mark)**

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d) The relative atomic mass of magnesium is 24.0. Calculate the enthalpy change, ΔH, for the reaction per mole of magnesium. **(1mark)**

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PROCEDURE II

i. Rinse the burette and fill it with 0.15M sodium carbonate, solution C.

ii. Transfer **ALL** the solutions in the beaker from procedure I into a 250ml volumetric flask. Rinse both the beaker and thermometer with distilled water and add the rinsing to the volumetric flask. Add more distilled water to the volumetric flask up to the mark. Label this as solution T.

iii. Pipette 25cm³ of solution T into a 250ml conical flask and add 2 drops of methyl orange indicator.

iv. Titrate solution T in the conical flask with the sodium carbonate, solution C and record the results in Table 2.

v. Repeat steps (iii) and (iv) and complete Table 2.

Table 2

Titre	1	2	3
Final burette reading (cm^3)			
Initial Burette reading (cm^3)			
Volume of solution C (cm^3)			

(4 marks)

a) Calculate the average volume of the 0.15M sodium carbonate, solution C used**.** **(1 mark)**

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b) Calculate the number of moles of :

i. Sodium carbonate used. **(1 mark)**

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ii. Hydrochloric acid in the 25.0 cm^3 of solution T. **(1 mark)**

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iii. Hydrochloric acid in the 250 cm^3 of solution T. **(1 mark)**

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iv. Hydrochloric acid that reacted with magnesium metal. **(1 mark)**

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v. Total number of moles of hydrochloric acid in the 50.0 cm^3, solution B. **(1 mark)**

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c) Determine the concentration of hydrochloric acid in moles per litre, in solution B. **(1 mark)**

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2. You are provided with solid P. Carry out the following tests and write your observation and inferences in the spaces provided.

Place all of the solid P in a boiling tube. Add about 10cm³ of distilled water and shake until all the solid dissolves. Divide the mixture into 6 portions

Observation	Inferences
(1 mark)	**(½ mark)**

a) To the first portion add 2 drops of sodium sulphate solution provided.

Observation	Inferences
(1 mark)	**(1mark)**

b) To the second portion, add 2M sodium hydroxide solution dropwise until in excess.

c)

Observation	Inferences
(1mark)	**(1mark)**

To the third portion, add 2M ammonia solution dropwise until in excess.

d)

Observation	Inferences
(1mark)	**(½ mark)**

To the fourth portion, add 2 - 3 drops of aqueous potassium iodide solution.

Observation	Inferences
(½ mark)	**(½ mark)**

e) To the fifth portion, add 2 - 3 drops of barium nitrate solution.

Observation	Inferences
(1mark)	**(1mark)**

f) To about 1cm³ of the solution in a test tube, add 2 - 3 drops of lead (II) nitrate solution.

Observation	Inferences
(½ mark)	**(½ mark)**

3. You are provided with liquid **F in** a boiling tube. Carry out the following tests. Write your observations and inferences in the spaces provided.

a) Place about 1cm^3 of liquid **F** on a watch glass. Introduce a burning splint above the solution on the watch glass.

Observation	Inferences
(1mark)	**(1mark)**

b) To the remaining liquid in the boiling tube, add about 10 cm3 of distilled water. Divide the mixture in to 3 portions.

Observation	Inferences
(1mark)	**(½ mark)**

(i) **T**o the first portion in a test tube, add two drops of acidified potassium manganate (VII)

Observation	Inferences
(1mark)	**(1mark)**

ii) To the second portion in a test tube, add two drops of acidified potassium dichromate (VI)

Observation	Inferences
(1mark)	**(1mark)**

ii) To the 3rd portion of 2cm^3 of solution **F** add a spatula of sodium carbonate provided.

Observation	Inferences
(½ mark)	**(1mark)**

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